



Analogue versus digital TV (Credit: www.encoremultimedia.com)

Cable TV came to India at the same time as satellite TV. In case of analogue cable TV, VSB modulation was used for the base band as in the case of terrestrial analogue TV, while RF carriers in the range of 40MHz to 600MHz were employed.

Digital TV

A video signal is generated by means of scanning the source picture at the frame rate of 25 per second to reduce flicker. Successive frames contain very similar information, so removal of redundant data helps save bandwidth.

In case of digital TV, pixel-wise data in each frame is stored, and redundant information is removed with the help of appropriate compression algorithm. The compressed digital video, formatted along with accompanied audio and other information, modulates a carrier in VHF, UHF or microwave bands, as per requirements of the transmission standard.

At the receiving end, demodulation and reconstruction of information is, at present, carried out in a unit called set-top box (STB). Subsequently, the reconstructed video and accompanied audio signals are interfaced to a regular TV receiver used as display device.

Digital satellite TV (DVB-S/S2)

Digital satellite TV broadcasting uses Digital Video Broadcasting - Satellite (DVB-S) and upgraded DVB-S2 standards. The video signal and accompanied audio are digitised, data compression is applied, and data packets modulate a carrier in Ka and Ku bands at 10GHz and 14GHz.

Single-carrier quadrature phase

shift keying (QPSK) is used for modulation of information. Signals are transmitted to ground by means of geostationary satellite in Earth's orbit and received by a 60cm parabolic reflector antenna looking towards the satellite. Antenna feed is integrated with a low-noise amplifier and a down converter to

convert RF signals in 10GHz - 14GHz band to a lower frequency of 950MHz - 2150MHz, called first intermediate frequency. This entire unit, called low-noise block (LNB), is mounted on the focal point of the parabolic dish.

The 950MHz - 2150MHz IF signal is brought down by means of a coaxial cable, to interface to an indoor STB. The latter down-converts this IF to a lower frequency, amplifies and demodulates it to recover the digital bit-stream. The latter is processed in the LSI device under program

DVB T2 Transmitters in India: Band: UHF, Modulation: QPSK, Bandwidth: 8MHz

City	Frequency (MHz)
Ahmedabad	562.00
Aurangabad	490.00
Bengaluru	498.00
Bhopal	490.00
Chennai	538.00
Cuttack	538.00
Delhi	578.00
Guwahati	538.00
Hyderabad	498.00
Indore	514.00
Jalandhar	570.00
Kolkata	482.00
Lucknow	562.00
Mumbai	474.00
Patna	498.00
Raipur	490.00
Ranchi	490.00
Srinagar	498.00
Thiruvananthapuram	530.00

(Source: Prasar Bharati website)

MECO
SINCE 1962

LOW COST LCD / LED MODULES



LC035



LC135

Specifications

3½ Digits, 1999 Counts (max.) LCD	3½ Digits, 1999 Counts (max.) LED
LC035	LC135
0 - 200mV DC 0 - 2, 20, 200V DC 0 - 200µA DC 0 - 2, 20, 200mA DC	
Auxiliary Power Supply	
9VDC ± 10%	5VDC ± 10%

Applications

- Ammeter
- Voltmeter
- Thermometer
- Power Supply
- PH Meter
- DB Meter
- Watt Meter
- LUX Meter
- LCR Meter
- Capacitance Meter
- Other Industrial and Domestic uses.

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